REJIVAN

"A Personal Nurse for Every Family"

Remote health monitoring · on-time medicines · automatic emergency help — in homes **and hospitals**, using wearables **plus privacy-first camera monitoring**, so families and staff can care for patients even when no one is in the room.

Concept Document — Working Draft v1.1 (for team planning & submission prep)

Competition	Hack for Social Cause (HSC) 2027 — Youth Tech Challenge
Organizers	Ministry of Youth Affairs & Sports (MoYAS) · Knowledge Partner: IIT Bombay
Thematic areas targeted	Healthcare, Wellbeing & Service Delivery + Elderly Care & Healthy Ageing
Motto	"Think Global, Hack Local — Technology for Social Transformation"
Use cases	**Home care** & **Hospital "Virtual Ward"** — rooms where doctors or nurses cannot always be present
Team	[Team name to be decided] — Team Lead, Developer, Content/Healthcare · State: Andaman & Nicobar Islands (UT)
Last date of submission	**15 October 2026 (Registration open: 1 Sep – 15 Oct 2026)**
National Showcase	VBYLD 2027, New Delhi — 10–12 January 2027
Status	Prototype v1 built — working draft (auto-updated 12 Sept 2026, 00:29)

1 Competition & Deadlines at a Glance

Item	Detail
Eligibility	Indian citizen; age 18–29 as on **17 August 2026**; enrolled in an **AISHE-registered** college/university/polytechnic; team of up to **3** (same or different institutions; solo allowed).
Registration process	1) Every member registers individually on mybharat.gov.in → 2) Team Lead registers the team under "Hack for Social Cause" → 3) Team lead uploads all submissions.
Submission window	**1 September – 15 October 2026**
What to submit	Problem Statement · 6–7 slide Presentation Deck (≤10 MB, incl. intelligence-tools disclosure) · Working Prototype (public GitHub/GitLab) · Demo Video 3–5 min, ≥720p, ≤80 MB (preferred) · Team Self-Declaration + student ID (Annexure 1).
Stages	Stage 1: Registration & submission (by 15 Oct) → Stage 2: State hackathon (16 Oct–30 Nov) → Stage 3: IIT Bombay screening to 36 national finalists (1–15 Dec) → Stage 4: National Showcase, New Delhi (10–12 Jan 2027).
Evaluation (6 parameters)	Relevance · Technical Strength · Functionality · Creativity/Innovation · Social Cause Impact · Presentation & Team.

Prizes	1st –25,000 · 2nd –50,000 · 3rd –25,000 · 4th/5th –15,000 each · National finalists also get a development grant of –6,000 + mentorship by IIT Bombay.

2 The Problem: Why This Matters Today

Millions of Indian families quietly struggle with the same fear: an elderly parent or a chronically ill family member (high blood pressure, diabetes, heart disease, post-stroke care) needs round-the-clock attention… but the family has to work, children study, and life moves to the city. The result is a daily chain of small disasters:

- **Medicines are missed or doubled** — average medication adherence in India is below 50% for chronic conditions, leading to hospital re-admissions and complications (WHO reports poor adherence as a leading cause of avoidable deaths).
- **Danger signs are noticed too late** — a heart-rate or blood-pressure shift at 2 a.m. goes unnoticed until the morning, when it is a crisis instead of a warning.
- **Hospitals are treated as the default answer** — patients are admitted, routinely monitored, and discharged, while the family can rarely continue safe monitoring at home. Out-of-pocket health spending pushes families into debt.
- **Professional monitoring is unaffordable** — remote-patient-monitoring (RPM) devices and private care plans cost anywhere from –20,000 to lakhs per year — out of reach for most Indian households.
- **Emergency response has no bridge** — by the time a family member dials 108 or an ambulance, crucial minutes are already lost; there is no automatic, trustworthy path from a dangerous reading to the nearest emergency service.
- **Hospitals are short-staffed too** — general wards and step-down units cannot station a nurse at every bed 24×7; patient falls and sudden deterioration happen inside rooms while no one is watching, and the hospital pays a heavy cost (falls are among the top hospital liabilities in India).
- **Rural and non-English-speaking India is left out** — most monitoring apps are English-only, need stable internet, and ignore the elderly themselves, who are often the least comfortable with smartphones.

Andaman & Nicobar Islands context (our "Hack Local")

The Andaman & Nicobar Islands — a Union Territory of 36 inhabited islands — concentrate every problem we solve in one place: a scattered, ageing population (a large retired/ex-servicemen community), a high burden of hypertension and diabetes, and only one major referral hospital (GB Pant Hospital, Port Blair) with specialists concentrated on the main island. Outer-island residents depend on Primary Health Centres and Cottage Hospitals, must travel by sea/air for specialist care, and face seasonal connectivity gaps. Families split between islands and the mainland, unreliable mobile networks beyond the bigger islands, and English-only apps are exactly the scenario our off-the-shelf, offline-friendly, multilingual design serves.

3 Problem Statement & Target Beneficiaries

> **Problem Statement (draft):** Families of elderly and chronically ill patients lack an affordable way to keep their loved ones safe and medically monitored between hospital visits. This leads to missed medicines, late recognition of danger, and avoidable emergencies. Existing monitoring solutions are costly, English-only, data-siloed, and do not automatically connect families with emergency services — leaving most Indian families dependent on expensive hospital care.

Beneficiary	How ReJivan helps
Elderly (60+) living alone or with working family	24×7 monitoring, medicine reminders, and automatic escalation if readings turn dangerous.
Chronically ill patients (BP, diabetes, heart, post-stroke)	Daily vitals tracking, trends visible to the doctor/family, early-warning “danger score”.

Family caregivers in cities & villages	See vitals and alerts from anywhere; confirm & act with one tap; no need to be physically present 24×7.
Community health workers / PHC staff	A simple dashboard and alert trail for outreach, home visits and follow-up.
State health authorities	Anonymised, privacy-safe trends for preventive-care planning (future phase).

4 Our Solution: ReJivan

ReJivan is **a personal Prajñā nurse that works in every home and every hospital ward** — an affordable, family-first and hospital-friendly health-monitoring system with these promises:

Promise	What it does
Monitor 24×7	An easy At-Home Monitor / wearable continuously tracks heart rate, blood oxygen (SpO^2), blood pressure and temperature — no nurse or hospital bed required on-site.
Camera watch (privacy-first)	Optional room cameras detect **falls, "out of bed", prolonged stillness and distress movements**. The intelligence works on the spot — **no video is recorded or stored**; only meaningful alerts come out. Consent-based, always.
Medicine on time	A smart medication manager reminds at the right time, confirms "taken/skipped", and notifies the family if a dose is missed — stopping a silent killer: missed medicines.
Prajñā that warns early	A transparent rule-based "Prajñā nurse" watches all readings, computes a clear **Safe / Caution / Danger** score, and explains in simple words why it is worried.
Family in the loop	A clean family dashboard on phone & computer shows live numbers, trends, alerts and medication status — in Bengali, Hindi and English, and readable even by non-technical users.
Auto emergency help	If a reading crosses danger limits: → family alerted first → if not confirmed/acknowledged → the system automatically escalates to ambulance/hospital/108 & helplines (e.g. tele-MANAS) — turning minutes into seconds.

Two monitoring layers that work together

Layer	What it measures	Where it runs
Wearable / At-Home Monitor	Heart rate, SpO^2, blood pressure, temperature, trends.	Home and hospital beds (simulated in prototype; low-cost hardware in roadmap).
Room cameras (CCTV-style)	**Behaviour & safety:** falls, getting out of bed unassisted, remaining too still, distress movements, wandering — and can estimate heart/respiration rate without touching the patient.	Home rooms and hospital wards via camera "zones".

ReJivan in the hospital — the "Virtual Ward"

Where a nurse cannot stand at every bed 24×7, ReJivan acts as the eyes in the room: a nurse-station view lists every bed/room with live vitals and camera-zone status; falls and out-of-bed events raise priority alerts instantly; and a sudden danger reading automatically calls the on-duty team (escalating to family for home mode). This directly relieves India's nursing shortage and makes wards safer — a strong institution-ready use case beyond homes.

How it works (demo flow)

- **1.** Register the patient, add family caregivers, medicines with times, and optional camera zones.
- **2.** The At-Home Monitor (prototype: simulation engine + manual entry; real device in roadmap) streams vitals.
- **3.** The Prajñā nurse scores every reading in real time, and camera zones watch the room.
- **4.** Medicine reminders fire on schedule; results go to the family.
- **5.** Any "Danger" reading or camera event (fall, out-of-bed) starts the escalation chain: family/nurse station → emergency contacts → services; every event forms a medical trail for the doctor.

5 Prototype Scope: What is Real vs Simulated (Honest Plan)

Judges value an honest, working demo over an over-claimed one. Our submission will clearly mark the boundary.

Feature	In working prototype	Notes
Dashboard (live vitals, trends, alerts)	**Fully real**	Responsive web app — works on phone + computer browsers.
Medication manager & on-time reminders	**Fully real**	Schedule creation, reminders, taken/skipped tracking, missed-dose alert to family.
Prajñā danger-score rules engine	**Fully real**	Transparent thresholds (BP, pulse, SpO^2, temp) — explainable, no black box.
Family dashboard & roles	**Fully real**	Multiple caregivers; per-role views; mobile-first.
Emergency escalation chain	**Fully real (flow)**	Alert family → Acknowledge → auto-call to configured emergency contacts; SMS/email/WhatsApp demonstration.
Camera-zone monitoring (fall / out-of-bed / low activity)	**Fully real (UI + event flow)**	Room "zones" with live simulated events + alerting; the underlying vision intelligence models (pose estimation, contactless vitals) are clearly documented and planned — not faked.
Hospital "Virtual Ward" mode	**Fully real (flow)**	Nurse-station view: bed/room list, live vitals, camera-zone status, priority alert queue — demonstrates staff relief in wards.
Signal "108 / ambulance" integration	Simulated	Simulated call ticket + escalation card; real integration is a pilot-phase item.
Vital sensing from hardware	Simulated	Built-in simulator generates realistic readings + manual entry; a low-cost ESP32 + pulse-oximeter hardware path is in the roadmap.
Payments / subscriptions	Simulated	Real plan-pages and pricing; actual payment gateway in pilot.
Languages	**Fully real (English + Bengali/Hindi labels)**	Vernacular-first UI for elderly/family.

Safety & ethics disclaimer (kept in app and submission): ReJivan is **not** a medical device, does not diagnose, and never replaces a doctor. It supports families and caregivers; all alerts offer a "human-in-the-loop" confirmation, and clinical validation is required before real deployment.

5b Live Prototype Status (auto-generated)

This section is built automatically from the code on every PDF rebuild — generated **12 Sept 2026, 00:29**. Feature status is read live from the prototype's transparency endpoint (`/api/simulation/status` in `prototype\server.js`), so it always matches what the app actually does.

Run the prototype: `cd prototype && npm start` → open `http://localhost:8080`.

Demo region: Andaman & Nicobar Islands (UT), India — homes in Port Blair + Little Andaman; Virtual Ward at GB Pant Hospital, Port Blair

Demo accounts (password: demo123)

Account	Login	Sees (data isolation)
Sharma Family	asharma@demo.in · password: demo123	Anita Sharma (home) + camera CAM1 + her medicines
Prakash Family	rprakash@demo.in · password: demo123	Ram Prakash (home) + camera CAM2 + his medicines
Ward Nurse Station	wardnurse@demo.in · password: demo123	Virtual Ward beds (Meera, Kavitha) + ward cameras

Feature status

Feature	Status	Notes
Login & per-user data isolation	Real	Scrypt-hashed passwords + session tokens; every account sees only its own patients, medicines, alerts and cameras
Dashboard (live vitals)	Real	Responsive web app, live refresh every 2 seconds, Safe / Caution / Danger badges
Medication manager	Real	Add schedule, mark taken, adherence log persisted to disk
Prajñā danger-score rules engine	Real	Transparent clinical thresholds for heart rate, SpO2, BP, temperature, glucose
Alert generation + escalation workflow	Real	Instant danger alerts with de-dupe cooldown and a full escalation log
Emergency auto-call chain	Real	Danger alert triggers an immediate call chain: family -> backup contact -> emergency services 108/112 (ambulance dispatch with GPS + vitals). Placement is SIMULATED (telecom API in production); the trigger/retry/escalation logic is real
Hospital Virtual Ward view	Real	Nurse-station bed list with live vitals and HIGH / MEDIUM / NORMAL priority queue
Privacy-first camera zones	Real	Fall / out-of-bed / low-activity event feed; on-device Prajñā, NO video recorded or stored
Live camera preview for family	Real	Privacy-safe real-time view of a camera zone (privacy-safe metadata feed rendered on canvas)
Multilingual UI	Real	English, Hindi, Bengali, Tamil, Telugu — including patient names
Reliability safeguards (7 layers)	Real	Data validation, confidence scoring, consecutive verification, sensor heartbeat, rate limiting, audit trail, graceful degradation — all wired into the alert pipeline

Medical device integration	Real	11 FDA/CDSCO-approved medical device profiles (SanketLife ECG, Omron BP, FreeStyle Libre 3 CGM, Biobeat chest patch, etc.); per-patient device registry with connection status, battery, signal strength; confidence scores boosted when medical devices are connected (80% vs 57% simulated); BLE heartbeat simulation; device catalogue with Indian-made options
Vital-sign data (patient readings)	Simulated	Simulation engine with per-condition profiles and danger episodes; when medical devices are connected, data quality improves to medical-grade confidence
Camera event data	Simulated	Event generator drives the fall / out-of-bed / low-activity alerts
SMS / WhatsApp / email / phone-call delivery	Simulated	Delivery endpoints are stubbed here; live APIs (telecom, WhatsApp) in production
Billing / payments	Simulated	Pricing pages planned; payment gateway in pilot phase

As reported by the running prototype (real vs simulated)

Layer status	Items
Real	authentication, data isolation per user, dashboard, medications, rules engine, alerts, escalation, emergency auto-call chain (priority + retry + escalation), multilingual UI, reliability safeguards (validation, confidence, consecutive verification, rate limiting, audit trail), medical device integration (CDSCO/FDA-approved device profiles, BLE connectivity simulation, per-patient device registry)
Simulated (honest disclosure)	vitals-data, camera-events, live-preview, billing, SMS/WhatsApp delivery, emergency phone calls

> **Prototype honesty:** simulated parts are clearly labelled in the app and this document. ReJivan is **not** a medical device, does not diagnose, and never replaces a doctor — a human caregiver or clinician always makes the final decision.

6 Technology Stack (v1 Prototype)

Layer	Choice	Why
Front-end	Responsive web app (HTML/CSS/JS) as a Progressive Web App	One build works on all phones/PCs, no app-store friction; judges click it live; later wrappable into an Android APK.
Back-end	Node.js + Express	Lightweight, easy to host and read for a student team.
Data	SQLite / JSON store	No server cost; portable; easy demo data.
Alerting	Email + SMS + WhatsApp APIs (demonstration endpoints)	Proves real escalation; free/low-cost tiers for demo.
Simulator	Built-in vitals simulator + manual entry	Realistic demo without hardware during evaluation.
Camera intelligence (vision)	Prototype: realistic event simulator; real path: on-device pose-estimation, fall detection & contactless-vitals models	Privacy-first: inference runs locally, no raw video stored or transmitted.
Repo & license	Public GitHub/GitLab · MIT	Meets HSC open-source + README/architecture/documentation requirement.

6.1 AI models in ReJivan (specific & named)

ReJivan's AI is built from **specific, published, verifiable models** — not a vague "machine learning" black box. Every model below runs **on-device (privacy-first)**, and the transparent rule engine (NEWS2-style) stays the permanent safety net: **a model never makes a life-or-death call alone**.

AI job in ReJivan	Named model / technique	What it adds	Status
Deterioration early-warning (vitals) — the "Prajnā nurse"	Rule engine aligned to hospital standard **NEWS2 / MEWS** thresholds (transparent, explainable) + an on-device **Isolation Forest** personal-baseline score	Flags readings that quietly drift away from *this patient's own normal*, ahead of fixed rules	**Real in prototype**
Vitals anomaly detection (time-series)	**LSTM autoencoder** + light alternative **ROCKET / MiniRocket**, trained per-patient on HR / SpO^2 / BP / temperature sequences	Catches "looks fine but is not fine for this patient" patterns; runs offline on the phone	Roadmap (demo-able with simulated data)
Universal signal option (research)	**UniTS** (CIKM '25 universal time-series foundation model) as a later single multi-signal scorer	One model reads HR + SpO^2 + BP + activity together	Roadmap
Fall detection (camera zones)	**Google BlazePose / MediaPipe Pose** (33 body keypoints only — no video) feeding a **LSTM–BiLSTM fall classifier** trained on public **UR-Fall / Le2i** datasets (published 95–99% accuracy)	Instant fall / collapse-to-floor recognition; nothing recorded	Roadmap (camera-zone UI already real)
Presence / out-of-bed (camera zones)	Lightweight on-device person detection + tracking (**YOLOv8**-class detector or MediaPipe detector)	Out-of-bed, prolonged stillness, wandering, empty-room detection	Roadmap
Contactless vitals (advanced)	**rPPG — remote photoplethysmography** (heart/respiration rate from the camera face region; MIT-Pulse line of research)	Vitals without touching the patient	Roadmap (lab-documented, not demoed)
Plain-language family assistant (optional)	Small on-device **Google Gemma** model or a cloud LLM (**Gemini** / GPT-family), clearly labelled	Explains scores in simple Bengali / Hindi / English — never diagnoses	Optional add-on (not required for core demo)
On-device AI runtime	**LiteRT (TensorFlow Lite)** + **MediaPipe Tasks** (+ ONNX Runtime Mobile)	Every model above runs on the phone, fully offline	**Strategy ready (native Android app)**

Honesty note: all models listed are public, reuse-friendly research used across medical-IoT projects. ReJivan never uploads raw video or raw health data for AI — inference happens on the phone. In the v1 prototype the rule engine is real; the model layer is a documented, blueprint-ready roadmap matching Section 5's Real vs Simulated table.

7 Sustainability & Business Model

- **Affordable subscription — "a Prajñā nurse for every family."** Proposed tiers: Family plan ~–99/month, Caregiver+ plan ~–199/month (multi-member, report sharing), Institutional (PHC/old-age homes) pricing on request — a fraction of today's –20,000+ RPM packages.
- **Public-health convergence:** designed to plug into Ayushman Bharat/ABHA digital health IDs, PHC follow-up workflows and state health missions (future phase), enabling possible empanelment rather than pure B2C.
- **Roadmap to hardware:** a "Made in India" low-cost monitor (ESP32 + pulse-oximeter + BP cuff) targeted under –5,000 for rural homes; the app is device-agnostic first.
- **Hospital & institutional channel (new):** a per-bed "Virtual Ward" service and camera-zone safety package for hospitals, old-age homes and nursing homes — a B2B revenue line beyond home subscriptions that also strengthens pilots and reference customers.

8 Extra Winning Points (our edge before we even build)

1. Multiple touchpoints — home and hospital together

Our entry squarely addresses **Healthcare, Wellbeing & Service Delivery** and **Elderly Care & Healthy Ageing**, and it does not stop at the home — it extends into **hospital virtual wards**, relieving India's nursing shortage. Intentional multi-theme positioning.

2. Digital-Public-Infrastructure alignment

Designed for ABHA integration, tele-MANAS (14416) mental-health linkage, and convergence with Ayushman Bharat PHC follow-up — the vision of "Viksit Bharat."

3. Digital Inclusion — not just a health app

Bengali/Hindi/English UI, a deliberately simple elderly-friendly design, offline-first behaviour, and **SMS/feature-phone fallback** for family alerts where internet is weak — this addresses the "last mile" the guidelines reward.

4. Privacy & trust as a feature (DPDP-aligned)

Explicit consent, patient-control of data, no ads, data minimisation, India-first storage plan — designed from day one to meet the Digital Personal Data Protection expectations judges increasingly probe.

5. Climate & community resilience angle

Heatwave/humidity-aware alerts (e.g. elderly during Bengal summers) tie the health solution to disaster-resilience too — a differentiator few teams will show.

6. Women-led & inclusive caregiving design

Designed around the reality that most unpaid caregiving is done by women — and the guidelines prefer a female team member & a non-tech member. Our content/video lead can be from a healthcare/social-science background.

7. Honesty that scores with judges

Clear "real vs simulated" boundaries, medical disclaimers, human-in-the-loop confirmation before any emergency action, and a full intelligence-usage disclosure (as rules require) — credibility is a differentiator.

8. Measurable impact story

Demo metrics: e.g. "adherence up from <50% to 90%", "response time to danger reading from ~hours to seconds", "cost of monitoring cut by >80% vs hospital."

9. SDG alignment

Explicitly mapped to UN SDG 3 (Good Health & Well-Being), SDG 9/10 (innovation & reduced inequalities) — shown on one slide.

10. Built to be open source

MIT-licensed, clean architecture, README + sample data — matching the repository requirements and encouraging state-level reuse.

11. Camera monitoring with privacy by design

Room cameras raise valid privacy fears, so we answer them head-on: on-device Prajñā, **zero video recording or storage**, alert-only output, explicit consent and DPDP alignment — turning the riskiest part of this idea into a trust selling point and a hospital-compliance advantage.

9 Mapping to the Evaluation Matrix

Evaluation parameter	How ReJivan answers it

Relevance	Direct match to two listed thematic areas; state-specific (Andaman & Nicobar Islands ageing + remote islands) problem statement.
Technical Strength	Clean responsive PWA + Node backend + documented rules engine + simulator; open-source MIT repo with README & architecture.
Functionality	Live clickable dashboard: vitals, medicines, alerts, escalation — demonstrated end-to-end in the video.
Creativity / Innovation	“Prajñā nurse at home” model: affordability + vernacular + SMS fallback + dual-theme positioning.
Social Cause Impact	Concrete metrics story (adherence, response time, cost); elderly + rural beneficiaries; SDG 3.
Presentation & Team	Role-based team (lead / dev / healthcare-content), 3–5 min narrative video, 6–7 slide deck with digital-tools disclosure.

10 Submission Checklist & Naming Convention

Files must be named **<State>_<TeamName>_<Document>.pdf** (e.g. AndamanNicobar_TeamX_ProblemStatement.pdf).

Deliverable	Specification	Status
Problem Statement	State-specific problem, beneficiaries, current gaps.	In progress
Presentation Deck (6–7 slides)	PPT/PDF, ≤10 MB; problem → solution → impact → team + digital-tools disclosure.	To build
Working Prototype	Public GitHub link: code, README, architecture, sample data, license.	To build
Demo Video (3–5 min)	MP4/MOV, ≥720p, ≤80 MB; team + problem + prototype + impact.	To build
Annexure 1 + IDs	Team details, signed self-cert, self-attested student IDs.	Pending

11 Working Timeline (from today, 08 Sep 2026)

Dates	Milestone
08–20 Sep	Finalise problem statement & MVP scope; build prototype v0 (dashboard + simulator + medicines).
20 Sep – 05 Oct	Add Prajñā danger engine + family roles + escalation flow; local-language labels; polish UI for demo.
05–10 Oct	Write README/architecture, sample data, push open-source repo; prepare 6–7 slide deck.
10–14 Oct	Record 3–5 min demo video (≥720p); collect Annexure-1 + ID proofs; final review.
On/before 15 Oct	**Team Lead submits everything on MY Bharat Portal.**
23–05 Nov	State-level hackathon & demonstration (mentoring + jury).
01–15 Dec	IIT Bombay screening to national finalists (36 teams).
10–12 Jan 2027	National Showcase, New Delhi — stall, pitch, awards.

12 Team & Roles (to finalise)

Role	Responsibility	Suggested member
Team Lead	Registration, submissions, deck, coordination.	[Name]
Developer	Prototype build, repo, code quality, demo pipeline.	[Name]
Healthcare / Content & Video	Problem statement, medical accuracy, video narration, local outreach.	[Name — ideal: female / social-science]

Open action items (next session)

- Decide final team name (appropriate, non-derogatory; e.g. “Sahaay Tech”).
- Confirm every member is individually registered on MY Bharat portal.
- Confirm AISHE college/institution details before 15 Oct.
- Approve final team name (working project title: **ReJivan**; team name TBD).

Digital-Tools Disclosure (required by rules): This concept document and the prototype are being developed with a digital coding assistant (opencode/CLI) under team supervision. The final submission & presentation deck will clearly disclose the extent of digital-tool usage, as mandated for HSC 2027.